

1. Electricity Bill Calculation — `if-else-if`

Problem Statement:

Write a Java program to calculate an electricity bill based on the number of units consumed.

Units	Rate
0–100	₹2 per unit
101–200	₹3 per unit
201–300	₹5 per unit
Above 300	₹7 per unit

Read the number of units in `main()`. Pass it to a static method `calculateBill()`. The method should calculate and **return the bill amount** to `main()`.

Concepts: `int`, `double`, arithmetic operator, `if-else-if`, static method, return value.

2. Student Grade Calculation — `if-else-if`

Problem Statement:

Write a Java program to calculate a student's grade based on marks.

Marks	Grade
90–100	A
75–89	B
60–74	C
50–59	D
Below 50	F

Read the marks in `main()`. Pass the marks to a static method `calculateGrade()`. The method should **return the grade** to `main()`.

Concepts: `int`, relational operators, `if-else-if`, `char/String`, static method.

3. Product Discount — `if-else-if`

Problem Statement:

Write a Java program to calculate the final price of a product after applying a discount.

Purchase Amount	Discount
₹10,000 and above	20%
₹5,000–₹9,999	10%

Purchase Amount	Discount
₹2,000–₹4,999	5%
Below ₹2,000	No discount

Read `productId`, `productName`, and `price` in `main()`. Pass the price to a static method `calculateFinalPrice()`. The method should calculate and **return the final price**.

Display the product details and final price in `main()`.

Concepts: `int`, `String`, `double`, arithmetic operators, `if-else-if`.

4. Largest of Two Numbers — `if-else`

Problem Statement:

Write a Java program to find the largest of two numbers.

Read two integers in `main()`. Pass both numbers to a static method `findLargest()`. The method should compare the numbers and **return the larger number** to `main()`.

If both numbers are equal, return either number.

Concepts: `int`, relational operator, `if-else`, static method, parameters, return.

5. Check Voting Eligibility — `if`

Problem Statement:

Write a Java program to check whether a person is eligible to vote.

Read the person's age in `main()`. Pass the age to a static method `isEligible()`. The method should return `true` if the age is **18 or above**, otherwise return `false`.

Display an appropriate message in `main()` based on the returned result.

Concepts: `int`, relational operator, `boolean`, `if`, static method.

6. Employee Salary Calculation — `if-else`

Problem Statement:

Write a Java program to calculate an employee's final salary.

Read the employee's basic salary in `main()`.

- If basic salary is ₹50,000 or above, add **20% HRA**.
- Otherwise, add **10% HRA**.

Pass the basic salary to a static method `calculateSalary()`. The method should calculate and **return the final salary**.

Concepts: `double`, arithmetic operators, `if-else`, percentage calculation, static method.

7. Temperature Conversion — Operators + `if`

Problem Statement:

Write a Java program to convert temperature from Celsius to Fahrenheit.

Formula:

$$\text{Fahrenheit} = (\text{Celsius} \times 9 / 5) + 32$$

Read the Celsius temperature in `main()`. Pass it to a static method `convertTemperature()`. The method should **return the Fahrenheit temperature**.

In `main()`, also check whether the converted temperature is above 100°F and display an appropriate message.

Concepts: `double`, arithmetic operators, `if`, static method, return value.

8. Simple Calculator — `if-else-if`

Problem Statement:

Write a Java program to perform a basic arithmetic operation.

Read two numbers and an operator (+, -, *, /) in `main()`.

Pass all three values to a static method `calculate()`.

The method should perform the requested operation and **return the result**.

For division, handle division by zero using a conditional statement.

Concepts: `double`, `char`, arithmetic operators, `if-else-if`, parameters, return value.

9. Bus Ticket Fare — `if-else-if`

Problem Statement:

Write a Java program to calculate a bus ticket fare based on passenger age.

Age	Fare
Below 5	Free
5–12	₹20
13–59	₹40
60 and above	₹25

Read the passenger's name and age in `main()`. Pass the age to a static method `calculateFare()`.

The method should **return the ticket fare** to `main()`.

Display the passenger name, age, and fare.

Concepts: `String`, `int`, `if-else-if`, relational operators, static method.

10. Mobile Data Usage — `if-else-if`

Problem Statement:

Write a Java program to calculate the mobile data charge based on data usage.

Data Used	Charge
Up to 1 GB	₹50
More than 1 GB up to 5 GB	₹100
More than 5 GB up to 10 GB	₹200
Above 10 GB	₹350

Read the customer's mobile number and data usage in GB in `main()`.

Pass the data usage to a static method `calculateCharge()`.

The method should calculate and **return the charge** to `main()`.

Display the mobile number, data usage, and final charge.

Concepts: `long`, `double`, arithmetic, `if-else-if`, static method, parameters, return.